

Sample Size and Power Calculations for Periodontal and Other Studies with Clustered Samples Using the Method of Generalized Estimating Equations

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Summary

In this paper we provide sample size or power estimation formulas for clinical trials that involve clustered samples and intend to use the method of Generalized Estimating Equations (GEE) for data analyses. The general formulas and two special cases, one for continuous and another for binary variables are given. The motivating example of periodontal clinical trial is given for illustration. These formulas should help researchers in designing studies with appropriate sample size and power.

Key words: Within-subject and between-subject variances; Intra-class correlation; GEE; Quasi-likelihood; Link function; ANOVA; Nested model; Periodontitis.

1. Introduction

Independence among the sample observations is one of the key assumptions underlying the application of many standard statistical methods. This assumption may be violated if multiple observations (subsamples) are taken on the same patient (cluster). In periodontal studies, for example, patients often have more than one tooth being examined. The patients are referred as clusters and the teeth (sites) are subsamples.

As pointed out by many authors (e.g., IMREY, 1986; DONNER and BANTING, 1988) there is strong evidence to indicate that tooth/site behavior cannot be treated as independent within patients. In particular, "the host factors influencing all sites within a mouth, viz., characteristics of immune responsiveness, nutrition, saliva production, composition and distribution, dietary and oral hygiene habits, microbiologic profile, and concomitant disease strongly suggest that sites within the same mouth will show association when properly monitored." Ignoring the positive correlation among tooth sites within patients may result in an under-estimate

of the variance, hence higher false positive rate. In planning for clinical trials, ignoring the positive correlation will likely underestimate the required sample size.

There have been many statistical methods suggested in the literature to handle clustered data an associated sample size estimation. In repeated measures context, for example, VONESH and SCHORK (1986) assumed normally distributed responses and estimated sample size based on the minimum correlation between, and the maximum variance of, the responses. ROCHON (1991) suggested use of Hotelling's T^2 to compare the vector of sample means between two treatment groups. OVERALL and DOYLE (1994) used the autoregressive (AR) model to characterize the correlation structure among the repeated measures. Discussions between Rochon and Overall & Doyle can be further found in their Letter to Editors (1995). In the periodontal area, for example, STERNE et al. (1988), MARKS (1989), KOCH and BECK (1989), and DEROUEN (1989) discussed several parametric linear models including variance components analyses. For continuous variables, normality was assumed; for binary data, logistic regression model was used in these papers. LEE and DUBIN (1994) considered a weighted estimate of the binary response probability from individual clusters by assuming an underlying continuous distribution for the individual clusters to allow for varying cluster sizes. Recently, IMREY (1992) provided very insightful discussion on design and analytic concepts for periodontal clinical trials. In his paper, clustered categorical data was the focus for analytic method, using either weighted-least-squares or modified Pearson χ^2 by DONNER and BANTING (1988). The weighted least-squares method was later studied by LIPSITZ and FITZMAURICE (1994).

Generalized estimating equations (GEE) has become a popular method for analyzing correlated data. It has been used for longitudinal data analyses (ZEGER and LIANG, 1986) and teratological experiments (LIANG and HANFELT, 1994). The advantages of GEE include its feasibility for both discrete and continuous data, and flexibility in the assumption of the correlation structure. It requires only the first and the second moments, not the full specification, of the distribution of the response. In a GEE model, both patient-level and tooth-level covariates can be incorporated. All these features are important and favorable to the analysis of periodontal data.

In this paper, we use properties of the GEE method to estimate the sample size and power. We first outline the general case and give the general formulas, then detail two commonly encountered special cases, i.e., the identity link for continuous responses and the logit link for binary responses. More detailed description of the GEE method can be found in the references (e.g. ZEGER and LIANG, 1986; LIANG and ZEGER, 1986). The application on periodontal studies is illustrated in Section 6.

2. Brief Review of Generalized Estimating Equations (GEE)

Notation for data: Consider the observation (y_{ij}, x_{ij}) for the i th patient and j th tooth site; $i = 1, \dots, N$, and $j = 1, \dots, n_i$. Here y_{ij} is the response variable and x_{ij}

is a $p \times 1$ vector of covariates (either patient-level or tooth-level). Let y_i be the $n_i \times 1$ vector $(y_{i1}, \dots, y_{ini})'$ and X_i be the $n_i \times p$ matrix $(x_{i1}, \dots, x_{ini})'$ for the i th patient.

Quasi-likelihood specification: Define μ_i to be the conditional expectation of y_i given the covariates X_i . Suppose that μ_i is linked with the covariates X_i by $\mu_i = h(X_i\beta)$, where β is a $p \times 1$ vector of parameters. In quasi-likelihood, the conditional variance of y_{ij} given X_i , v_{ij} , is expressed as a known function of the expectation by $v_{ij} = g(\mu_{ij})/\phi$, where ϕ is a scale parameter.

GEE: A key step in GEE is to decompose and express the $(n_i \times n_i)$ conditional variance/covariance matrix of y_i given X_i , V_i , in terms of a 'working' correlation matrix $R_i(\alpha)$, which can differ from subject to subject, but is assumed to be fully specified by the $s \times 1$ vector of unknown parameters α : $V_i = A_i^{1/2} R_i(\alpha) A_i^{1/2} / \phi$, where A_i is an $(n_i \times n_i)$ diagonal matrix with $g(\mu_{ij})$ as the j th diagonal element.

The GEE estimate of β is the solution of the following equation:

$$\sum_{i=1}^N D_i' V_i^{-1} (y_i - \mu_i) = 0 \quad (1)$$

where $D_i = d\mu_i/d\beta$ is the $n_i \times p$ gradient matrix for the i th patient. Notice that the above equation involves both parameters α and β and, generally, needs to be solved iteratively.

Let $\hat{\beta}$ be the solution of (1). LIANG and ZEGER (1986) has shown that $N^{1/2}(\hat{\beta} - \beta)$ is asymptotically multivariate normal with covaricance matrix:

$$V_R = \lim_{N \rightarrow \infty} N(\Sigma_1^{-1} \Sigma_0 \Sigma_1^{-1}) \quad (2)$$

where

$$\Sigma_1 = \sum_{i=1}^N D_i' V_i^{-1} D_i \quad \text{and} \quad \Sigma_0 = \sum_{i=1}^N D_i' V_i^{-1} \text{cov}(y_i | X_i) V_i^{-1} D_i.$$

3. Sample Size and Power Calculations Using GEE –

The General Two-Sample Case

Suppose that we are interested in testing the treatment effect and that the treatment assignment for each patient is coded in the last column of the $(n_i \times p)$ covariate matrix X_i . The corresponding last parameter of β is β_p . The hypotheses of interest are $H_0 : \beta_p = 0$ versus $H_A : \beta_p = b > 0$.

A two sided test based on $\hat{\beta}$ with type-I error rate α and power $= 1 - \beta$, will then require the total sample size

$$N = \frac{v_R^* (Z_{\alpha/2} - Z_{1-\beta})^2}{b^2} \quad (3)$$

where v_R^* is the (p, p) -th lower corner element of V_R , and $Z_a = \Phi^{-1}(1 - a)$ is the $100(1 - a)\%$ percentile of the standard normal distribution.

Given the sample size N , the power is

$$1 - \beta = 1 - \Phi \left(Z_{\alpha/2} - \frac{b \sqrt{N}}{v_R^{*1/2}} \right). \quad (4)$$

Notice that we can simplify V_R in (2) for the sample size and power calculations. For data analysis, V_i is the working covariance matrix of y_i given X_i , and differs from the true $\text{cov}(y_i | X_i)$. For initial sample size or power calculations at the design stage however, we usually have to assume values for the variance and covariance of y_i . Hence V_i is the same as $\text{cov}(y_i | X_i)$ here. (Discussion of the sample size re-estimation strategy will be given later). Therefore $\Sigma_0 = \Sigma_1$ and $V_R = \lim_{N \rightarrow \infty} N(\Sigma_1^{-1})$. Further simplification of V_R is required for a specific application, since V_R still involves N and other parameters that require specification. This is demonstrated in two special cases as follows.

4. Special Case I – Identity Link Function

This section gives details for the case when $\mu_i = h(X_i\beta) = X_i\beta$ and $v_{ij} = g(\mu_{ij})/\varphi = \sigma^2$. At the design stage, for simplicity, we may assume a common number of tooth sites (subsamples) for each patient (i.e., $n_i = n$ for all $i = 1, \dots, N$) and that the patient allocations in the control and test treatment groups are, respectively, $N_1 = N\pi$ and $N_2 = N(1 - \pi)$. We may also assume that patients in the study have similar tooth sites (subsamples) in examination so that the correlation matrix $R_i(\alpha)$ is independent of i .

Furthermore, we let the covariate matrix X_i involve only two columns, a constant column of ones and a column indicating treatment group assignment (0 for control and 1 for the test group). Hence $\beta = (\beta_0, \beta_1)'$, where β_0 represents the expected response for a patient in the control group, and $\beta_0 + \beta_1$ is the expected response for patients in the test group. The hypothesis will then be $H_0 : \beta_1 = 0$ versus $H_A : \beta_1 = b > 0$.

With the above setting, $D_i = d\mu_i/d\beta = X_i$. Let X_c be the $n \times 2$ covariate matrix X_i for the N_1 patients in the control group. (Hence $X_c = [\mathbf{1}, \mathbf{0}]$, with ones in the first column and zeros in the second). Similarly, let X_t be the $n \times 2$ covariate matrix X_i for the N_2 patients in the test group. (Hence $X_t = [\mathbf{1}, \mathbf{1}]$, with ones in the first and the second columns.) Let $V_i = \sigma^2 R$ and R is the $n \times n$ correlation matrix for all $i = 1, \dots, N$. Thus,

$$\Sigma_1 = \sum_{i=1}^N D_i' V_i^{-1} D_i = N[(\pi X_c' R^{-1} X_c) + (1 - \pi) X_t' R^{-1} X_t)]/\sigma^2.$$

$$\begin{aligned}
V_R &= \lim_{N \rightarrow \infty} N(\Sigma_1^{-1}) = \sigma^2[(\pi X_c' R^{-1} X_c) + (1 - \pi) X_t' R^{-1} X_t]^{-1} \\
&= \sigma^2 \left[\begin{bmatrix} 1 & 1 - \pi \\ 1 - \pi & 1 - \pi \end{bmatrix} (\mathbf{1}' R^{-1} \mathbf{1}) \right]^{-1} \\
&= (\sigma^2 / \mathbf{1}' R^{-1} \mathbf{1}) \frac{1}{\pi(1 - \pi)} \begin{bmatrix} 1 - \pi & -1 + \pi \\ -1 + \pi & 1 \end{bmatrix}
\end{aligned}$$

and

$$v_R^* = \sigma^2 / [\mathbf{1}' R^{-1} \mathbf{1} \pi(1 - \pi)]. \quad (5)$$

If we assume correlations are exchangeable among the sites ($= \rho$), then the α parameter in R is just the scalar ρ . $\mathbf{1}' R^{-1} \mathbf{1} = \frac{n}{1 + (n - 1) \rho}$ and Eq. (5) becomes

$$v_R^* = \frac{\sigma^2 [1 + (n - 1) \rho]}{\pi(1 - \pi) n}. \quad (6)$$

Substituting (6) to the sample size formula (3), we have

$$N = \frac{\sigma^2 [1 + (n - 1) \rho] (Z_{\alpha/2} - Z_{1-\beta})^2}{\pi(1 - \pi) n b^2}. \quad (7)$$

Compared to the case of independent subsamples, (7) has the extra multiplier, $1 + (n - 1) \rho$, known as the 'design effect' factor in the sample survey literature (e.g., KISH, 1965). When $\rho > 0$, which we expect in the periodontal studies, the required number of patients is larger than the case of independent subsamples. A similar formula is given in DIGGLE et al. (1994, p. 31) for sample size per group with $\pi = 0.5$ and one sided tests.

When estimating sample size for a new study, we normally have to obtain some preliminary estimate of the parameters from previous studies, as shall be demonstrated in the example in Section 6. These previous studies are usually reported by the analysis of variance (ANOVA) with a two-stage nested model format. The within-subject (among the tooth sites) variance, σ_e^2 , and the between-subject variance, σ_b^2 , are useful information that are familiar to many researches (e.g., STERNE et al. 1988). It can be easily shown that the two-stage nested ANOVA model is a special case for which the within-subject, among-sites correlation is exchangeable and is equal to $\rho = \sigma_b^2 / (\sigma_b^2 + \sigma_e^2)$. If one also recognizes that the scale parameter σ^2 in GEE is the total variance, $\sigma_b^2 + \sigma_e^2$, then in formula (7) the quantity $\sigma^2 [1 + (n - 1) \rho] = \sigma_e^2 + n\sigma_b^2$, which is the expected mean squares in the two-stage nested ANOVA for testing the treatment effect. Furthermore, let $\bar{Y}_A$ and $\bar{Y}_B$ be the overall mean of the two treatment groups. It can be shown that $\text{Var}(\bar{Y}_A - \bar{Y}_B) = \frac{\sigma_e^2 + n\sigma_b^2}{N\pi(1 - \pi)n} = \frac{\sigma^2 [1 + (n - 1) \rho]}{N\pi(1 - \pi)n}$. Therefore, in this special case formula (7) is equivalent to the usual sample size formula for testing two-sample means of normally distributed variables with the appropriate variance substituted.

5. Special Case II – Binary Response

For the binary response, we use the logit model: $\mu_i = p_i = h(X_i\beta) = \exp(X_i\beta)/(1 + \exp(X_i\beta))$ and $v_{ij} = g(\mu_{ij})/\varphi = g(p_{ij}) = p_{ij}(1 - p_{ij})$. At the design stage, similar to the identity link case, we may assume, for simplicity, a common number of tooth sites for each patient, (i.e., $n_i = n$ for all $i = 1, \dots, N$) and that the patient allocations in the control and test treatment groups are, respectively, $N_1 = N\pi$ and $N_2 = N(1 - \pi)$. We may also assume that patients in the study have similar tooth sites (subsamples) in examination so that the correlation matrix $R_i(\alpha)$ is independent of i .

Again, consider the case where the covariate matrix X_i involves only two columns, a constant column of ones and a column indicating treatment group assignment (0 for control and 1 for the test group). Let $X_i = X_c = [1, 0]$, the $n \times 2$ covariate matrix for the N_1 patients in the control group and $X_i = X_t = [1, 1]$, the $n \times 2$ covariate matrix for the N_2 patients in the test group. Hence $\beta = (\beta_0, \beta_1)'$, and $p_{ij} = P_0 = \exp(\beta_0)/(1 + \exp(\beta_0))$, for all i in the control group and $j = 1, \dots, n$; and $p_{ij} = P_1 = \exp(\beta_0 + \beta_1)/(1 + \exp(\beta_0 + \beta_1))$, for all i in the test group and $j = 1, \dots, n$. Therefore, $\beta_1 = \log(P_1/(1 - P_1)) - \log(P_0/(1 - P_0))$ is the log (odds ratio) of the response probabilities. The null hypothesis of $P_0 = P_1$ is equivalent to $H_0 : \beta_1 = 0$, and the alternative hypothesis in term of log (odds ratio) is $H_A : \beta_1 = b > 0$.

With the above setting, $D_i = d\mu_i/d\beta = P_0(1 - P_0)[1, 0]$, for the N_1 patients in the control group, $D_i = P_1(1 - P_1)[1, 1]$, for the N_2 patients in the test group; and $V_i = P_i(1 - P_i)R$. Thus,

$$\begin{aligned}\Sigma_1 &= \sum_{i=1}^N D_i' V_i^{-1} D_i \\ &= N[\pi X_c' P_0(1 - P_0) R^{-1} X_c + (1 - \pi) X_t' P_1(1 - P_1) R^{-1} X_t] \\ &= N[\pi P_0(1 - P_0) [1, 0]' R^{-1} [1, 0] \\ &\quad + (1 - \pi) P_1(1 - P_1) [1, 1]' R^{-1} [1, 1]] \\ &= N(1'R^{-1}1) \begin{bmatrix} \pi P_0(1 - P_0) + (1 - \pi) P_1(1 - P_1) & (1 - \pi) P_1(1 - P_1) \\ (1 - \pi) P_1(1 - P_1) & (1 - \pi) P_1(1 - P_1) \end{bmatrix} \\ V_R &= \lim_{N \rightarrow \infty} N(\Sigma_1^{-1}) \\ &= (1'R^{-1}1)^{-1} \frac{1}{\pi P_0(1 - P_0)} \begin{bmatrix} 1 & 1 \\ -1 & \frac{\pi P_0(1 - P_0)}{(1 - \pi) P_1(1 - P_1)} + 1 \end{bmatrix}\end{aligned}$$

and

$$v_R^* = (1/1'R^{-1}1) \left[\frac{1}{\pi P_0(1 - P_0)} + \frac{1}{(1 - \pi) P_1(1 - P_1)} \right]. \quad (8)$$

Substituting (8) to the sample size formula (3), we have

$$N = 1/(\mathbf{1}'\mathbf{R}^{-1}\mathbf{1}) \left[\frac{1}{\pi P_0(1-P_0)} + \frac{1}{(1-\pi)P_1(1-P_1)} \right] \times \frac{(Z_{\alpha/2} - Z_{1-\beta})^2}{b^2}. \quad (9)$$

If we assume correlations are exchangeable among the sites ($= \rho$), then $\mathbf{1}'\mathbf{R}^{-1}\mathbf{1} = \frac{n}{1 + (n-1)\rho}$. Eq. (9) becomes

$$N = \frac{[1 + (n-1)\rho] (Z_{\alpha/2} - Z_{1-\beta})^2}{nb^2} \left[\frac{1}{\pi P_0(1-P_0)} + \frac{1}{(1-\pi)P_1(1-P_1)} \right] \quad (10)$$

where b is the log (odds ratio) $= \log [P_1(1-P_0)/(P_0(1-P_1))]$. Again, we see the 'design effect' factor, $1 + (n-1)\rho$, in (10).

DIGGLE et al. (1994, p. 31) also gave a formula of sample size per group, assuming $\pi = 0.5$, for one-sided tests with difference of P_0 and P_1 as the parameter of interest. LIANG and PULVER (1996) extended DIGGLE et al. (1994) by allowing different correlation, ρ , and sample size in each treatment group.

6. Example

A randomized study is designed to compare a new treatment with the standard treatment in patients with periodontitis. One of the clinical variable for efficacy is the periodontal ligamentous attachment level (AL). Suppose that in pilot studies

Table 1
Sample Size Estimation For Attachment Level (AL)

Detectable Mean Difference in AL- Gain (mm)	Tooth-sites Correlation	Number of Patients/Group (Assume average of $n = 4$ teeth per patient)			
		Total Variance AL Change $\sigma^2 = 1.5 \text{ mm}^2$		Total Variance AL Change $\sigma^2 = 1.0 \text{ mm}^2$	
		power 80%	power 90%	power 80%	power 90%
b	ρ				
0.10	0.2	942	1261	628	841
0.15	0.2	419	561	280	374
0.20	0.2	236	316	157	211
0.10	0.4	1296	1734	864	1156
0.15	0.4	576	771	384	514
0.20	0.4	324	434	216	289

the difference of the average gain in the attachment level between the treatment groups was found to be between 0.10 to 0.20 mm. The total variances (within patients-between tooth sites plus between patients) was about 1.0 to 1.5 mm². The correlation among tooth sites was about 0.2 to 0.4. Using these information, and assume an average of 4 teeth per patients in the study, Table 1 displays the estimated number of patients per group ($\pi = 0.5$) required to have a power of 80–90% to detect the corresponding mean difference in AL gain between groups at $\alpha = 0.05$ (two-sided).

In view of Table 1 with published periodontal articles, we find that many periodontal studies are under powered to detect a meaningful treatment effect on the ligamentous attachment level. To have a reasonable power for a moderate size of clinical trial, new clinical technologies are needed to help improving the precision of the probing method in measuring the attachment level (GROSSI et al., 1996). Pilot studies with small sample sizes should avoid using GEE since it is a highly asymptotic method.

7. Discussion

Motivated by the need of designing a proper periodontal study, this paper has worked out the sample size and power estimation using the GEE method. The special cases of identity link for continuous response and the logit link for the binary response are presented in detail. The explicit formulas are given when the site-to-site correlation are the same (i.e. the compound symmetry case). Compared to the formulas for the independent samples, we have seen the so-called 'design effect' factor appeared in both formulas for the clustered data. These special cases appear commonly in dental studies. However, it is also helpful to see that the general formulas (Eqs. 3 and 4) are available for other situations. Sample sizes with the general pattern of X_i which includes other covariates than treatment groups can readily be formulated by following the same steps. When planning studies, the incorporation of covariates should always be considered carefully, as the variance of the treatment difference may considerably change if the covariates are correlated with treatment.

The initial sample size and power estimation in the two special cases are made easy by assuming the exact variance-covariance values in the GEE at the design stage. As in the usual case of independent samples, these assumptions of variances and covariances need careful verification and may need to be re-estimated at an interim stage of the study. A re-estimation procedure in this case is conceivably more interesting than the independent samples case (e.g., SHIH, 1992; SHIH and ZHAO, 1997), since there is an extra site-to-site correlation to be estimated. When data at an interim stage become available, one can update the variance-covariance matrix and the average number of subsamples, and re-estimate of the sample size required. The work in SHIH and GOULD (1995) is closely related to this situation.

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